EXPERIMENT 12

Earley Parser for a Context-Free Grammar

Aim

To use an Earley parser to determine whether a sentence is accepted by a context-free grammar.

Procedure

1. Install NLTK if required.
2. Define a context-free grammar.
3. Create an Earley chart parser.
4. Parse the sample sentence.
5. Display whether a parse is found.

Program

```
# Install first: pip install nltk
from nltk import CFG
from nltk.parse import EarleyChartParser

grammar = CFG.fromstring("""
S -> NP VP
NP -> Det N
VP -> V
Det -> 'the'
N -> 'cat' | 'dog'
V -> 'sleeps' | 'runs'
""")
parser = EarleyChartParser(grammar)
sentence = "the cat sleeps".split()
trees = list(parser.parse(sentence))
print("Parse found" if trees else "No parse found")
if trees:
    print(trees[0])
```

Output

```
OUTPUT — EXPERIMENT 12
Parse found
(S (NP (Det the) (N cat)) (VP (V sleeps)))
```

Result

Thus, the Earley parser successfully found a parse for the sentence.